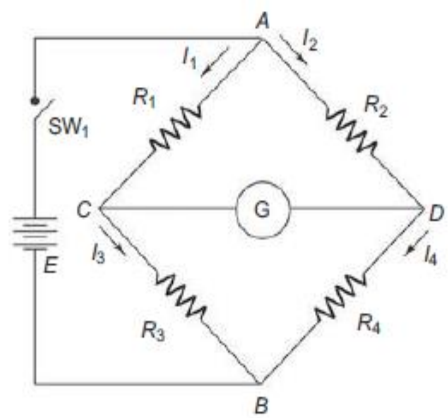
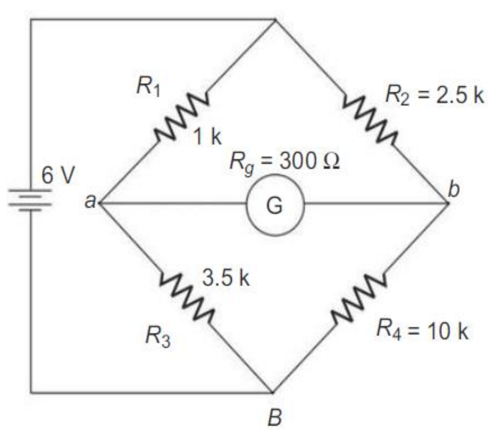
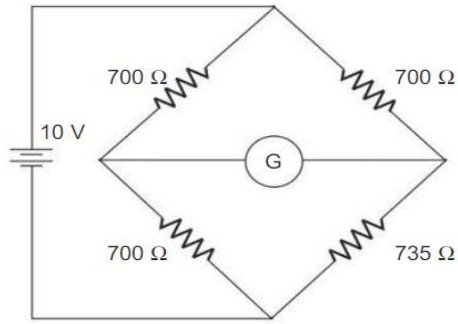
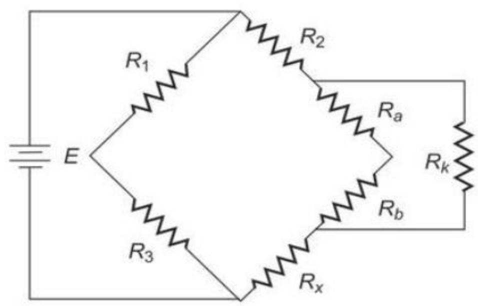
	GLOBAL ACADEMY OF TECHNOLOGY, RR NAGAR, BENGALURU-560 098	
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING		
SUBJECT: SENSORS AND INSTRUMENTATION		**SUBJECT CODE: BEC24406D**
QUESTION BANK		
Q. No.	**MODULE 1: Unit 1**	**Marks**
1	What is a System? What are the functions and objectives of Measurement System?	**7**
2	What are the advantages of Electronic Measurement Systems?	**6**
3	Classify Sensors according to Different Exhaustive Criteria.	**7**
4	With a neat diagram, explain Liquid-column U-tube manometer, C-shaped Bourdon tube, and Twisted Bourdon tube Pressure Sensor.	**7**
5	With a neat block diagram, explain the functions and data flow of a measurement and control system.	**7**
6	Write a note on Sensors, Transducers and Actuators.	**6**
7	With a neat diagram, explain Membrane diaphragm, Micromachined diaphragm, Capsule and Bellows Pressure Sensor.	**7**
8	Explain the different types of pressure sensors and mention the applications of pressure sensor.	**8**
Q. No.	**MODULE 1: Unit 2**	**Marks**
1	State Archimedes' buoyancy principle. Using the Archimedes' principle, explain float type level sensor.	**6**
2	Explain Temperature Sensors with a neat diagram.	**6**
3	List out the applications of Temperature sensors.	**6**
4	List out the applications of level sensors.	**6**
5	Explain the types of level sensors.	**6**
Q. No.	**MODULE 1: Unit 3**	**Marks**
1	Write a note on Material for sensors.	**7**
2	Explain thick film technology.	**7**
3	Illustrate Micro sensor Technology.	**7**
4	Differentiate between thick film and thin film technology.	**6**
5	Explain thin film technology.	**7**
6	Explain micromachining technologies.	**7**
7	Differentiate between micromachining technologies and Micro sensor.	**6**

Q. No.	MODULE 2: Unit 1	Marks
1	With a neat circuit diagram, explain the working of ideal potentiometer connected to voltmeter.	**10**
2	Explain tilt sensors based on potentiometer.	**10**
3	Explain fundamentals involved in strain Guage.	**10**
4	Discuss Thermistors in detail.	**10**
5	Discuss RTD's in detail.	**10**
6	A given PRT probe has 100 Ω and $\alpha = 0.00389$ (Ω/Ω)/K at 0 °C. Calculate its sensitivity and temperature coefficient at 25 °C and 50 °C.	**10**
7	A model of $R_T = Ae^{B/T}$. Determine A for a unit having B = 4200 K and 100 kΩ at 25 °C. Calculate the value for α at 0 °C and 100 °C.	**10**
8	Calculate B for an NTC thermistor that has 5000 Ω at 25 °C and 1244 Ω at 60°C.	**10**
9	Discuss Resistive Temperature Detector (RTD) with symbols and explain the catalytic gas sensor application of RTD.	**10**
10	List out the thermistor types and its application.	**10**
Q. No.	**MODULE 2: Unit 2**	**Marks**
1	A common sensor for traffic control is a buried flat coil supplied by a current from 20kHz to 150kHz. As a car enters the coil, its inductance decreases because of eddy currents induced in the car, and when the car leaves the coil, the inductance returns to its original value. If the coil is included in a harmonic oscillator, the change in oscillation frequency signals the presence of a car. If the oscillation frequency changes by 10% because of a car and we measure it with an 8-bit counter, determine the maximal measurement time before the counter overflows.	**10**
2	Explain sensors based on Quartz Resonator.	**10**
3	Explain the block diagram of Digital Quartz Thermometer.	**10**
4	With a neat block diagram, explain basic frequency meter.	**10**
5	With a neat block diagram, explain period meter.	**10**
6	A given sensor has an output frequency of 9kHz to 11kHz. Determine the measurement time and the number of counts needed to measure the frequency with a 12-bit resolution.	**10**
7	A given sensor has an output frequency of 9kHz to 11kHz. Determine how many signal cycles must elapse to measure their period with 12-bit resolution by using an 8051 microcontroller whose clock frequency is 12MHz.	**10**
8	Differentiate between frequency meter and period meter.	**10**
9	A given sensor has an output frequency from 9KHz to 11KHz. How many signal cycles must elapse to determine their frequency with a 12-bit resolution from the measurement of its period using an 8051 Microcontroller whose clock frequency is 12MHz. What is the resolution needed for the period measuring and how long does the measurement take?	**10**
10	Mention the applications of resonant sensors.	**10**

<table>
<tr><th>Q. No.</th><th>MODULE 3: Unit 1</th><th>Marks</th></tr>
<tr><td>1</td><td>Explain the static characteristics of an instrument.</td><td>10</td></tr>
<tr><td>2</td><td>Explain the different types of static errors.</td><td>10</td></tr>
<tr><td>3</td><td>The expected value of the voltage across a resistor is 80 V. However, the measurement gives a value of 79 V. Calculate (i) absolute error (ii) % error (iii) relative accuracy (iv) % of accuracy.</td><td>10</td></tr>
<tr><td>4</td><td>Table gives the set of 10 measurements that were recorded in the laboratory. Calculate the precision of the 6th measurement.
<table>
<tr><td>Measurement Number</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr>
<tr><td>Measurement Value</td><td>101</td><td>102</td><td>97</td><td>101</td><td>100</td><td>103</td><td>98</td><td>106</td><td>99</td></tr>
</table></td><td>10</td></tr>
<tr><td>5</td><td>The expected value of the voltage across a resistor is 90 V. However, the measurement gives a value of 89 V. Calculate (i) absolute error (ii) % error (iii) relative accuracy (iv) % of accuracy.</td><td>10</td></tr>
<tr><td>6</td><td>Write a note on Error in measurement and list out the sources of error.</td><td>10</td></tr>
<tr><td>7</td><td>The expected value of the current through a resistor is 20 mA. However, the measurement yields a current value of 18 mA. Calculate (i) absolute error (ii) % error (iii) relative accuracy (iv) % accuracy.</td><td>10</td></tr>
<tr><td>8</td><td>A voltmeter having a sensitivity of 1kΩ/V is connected across an unknown resistance in series with a milliammeter reading 80V on 150V scale. When the milliammeter reads 10mA, calculate the (i) Apparent resistance of the unknown resistance (ii) Actual resistance of the unknown resistance (iii) Error due to the loading effect of the voltmeter.</td><td>10</td></tr>
<tr><td>9</td><td>If the milliammeter reads 600 mA and the voltmeter reads 30 V on a 150 V scale, calculate the following: (i) Apparent, resistance of the unknown resistance (ii) Actual resistance of the unknown resistance (iii) Error due to loading effect of the voltmeter Comment on the loading effect due to the voltmeter.
Given Voltmeter sensitivity = 1000 W/V.</td><td>10</td></tr>
<tr><td>10</td><td>What are digital voltmeters? Mention the features, operating and performance characteristics of Digital voltmeter.</td><td>10</td></tr>
<tr><th>Q. No.</th><th>MODULE 3: Unit 2</th><th>Marks</th></tr>
<tr><td>1</td><td>Explain the operation of Dual slope integrating type DVM with diagram.</td><td>10</td></tr>
<tr><td>2</td><td>Describe the operation of Successive approximation DVM with diagram.</td><td>10</td></tr>
<tr><td>3</td><td>Explain digital tachometer with a neat block diagram.</td><td>10</td></tr>
<tr><td>4</td><td>Explain digital pH meter with a neat block diagram.</td><td>10</td></tr>
<tr><td>5</td><td>Explain digital phase meter with a neat block diagram.</td><td>10</td></tr>
<tr><td>6</td><td>Explain the operation of Ramp type DVM with diagram.</td><td>10</td></tr>
<tr><td>7</td><td>An integrator contains 100kΩ and 1μF capacitor. If the voltage applied to the integrator output is 1V, what voltage will be present at the output of the integrator after 1s? If a reference voltage is applied to the integrator, at time t1 is 5V in amplitude, what is the time interval of t_2?</td><td>10</td></tr>
</table>

8	Suppose the converter can measure a maximum of 5V, i.e. 5V corresponds to the maximum count of 11111111. If the test voltage V_{in}=1V, explain the steps that will take place in the measurement along with various output levels for each bit.	**10**
9	List out the applications of digital tachometer, digital pH meter and digital phase meter.	**10**
10	Explain the basic principle of dual slope type DVM and mention the applications of dual slope type DVM.	**10**

Q. No.	MODULE 4: Unit 1	Marks
1	Explain the operation of balanced Wheatstone's bridge and derive the expression for unknown Resistance in a balanced Wheatstone's bridge.	**10**
2	Explain the operation of unbalanced Wheatstone's Bridge.	**10**
3	Explain the operation of slightly Unbalanced Wheatstone's Bridge	**10**
4	Mention the Applications and Limitations of Wheatstone's Bridge.	**10**
5	Describe the operation of Kelvin's bridge and derive the expression for unknown resistance in Kelvin's bridge.	**10**
6	Figure consists of the following parameters. R_1 = 10 kΩ, R_2 = 15 kΩ and R_3 = 40 kΩ. Find the unknown resistance R_4. 	**10**
7	An unbalanced Wheatstone bridge is given in Figure. Calculate the current through the galvanometer. 	**10**

8	Given a center zero 200–0–200mA movement having an internal resistance of 125Ω. Calculate the current through the galvanometer given in Figure by the approximation method.  	**10**
9	If in Figure, the ratio of R_a to R_b is 1000Ω, R_1 is 5Ω and $R_1=0.5R_2$. What is the value of R_x?  	**10**
10	Mention the applications of Kelvin's bridge.	**10**
Q. No.	**MODULE 4: Unit 2**	**Marks**
1	Explain the operation of Capacitance Comparison bridge.	**10**
2	Explain the operation of Inductance Comparison bridge.	**10**
3	Explain the operation of Maxwell's bridge with circuit diagram.	**10**
4	Describe the operation of Wien's bridge with circuit diagram.	**10**
5	A capacitance comparison bridge is used to measure a capacitive impedance at a frequency of 2kHz. The bridge constants at balance are $C_3=100\mu F$, $R_1=10k\Omega$, $R_2=50k\Omega$, $R_3=100k\Omega$. Find the equivalent series circuit of the unknown impedance.	**10**
6	An Inductance comparison bridge is used to measure inductive impedance at a frequency of 5kHz. The bridge constants at balance are $L_3=10mH$, $R_1=10K\Omega$, $R_2=40K\Omega$ and $R_3=100K\Omega$. Calculate equivalent series circuit of unknown impedance.	**10**
7	A Maxwell's bridge is used to measure an inductive impedance. The bridge constants at balance are $C_1=0.01\mu F$, $R_1=470k\Omega$, $R_2=5.1k\Omega$ and $R_3=100k\Omega$. Find the series equivalent of the unknown impedance.	**10**
8	A Wein bridge circuit consists of the following: $R_1=4.7k\Omega$, $R_2=20k\Omega$, $R_3=10k\Omega$, $R_4=100k\Omega$, $C_1=5nf$, $C_3=10nf$ Determine the frequency of the circuit.	**10**

9	Find the equivalent parallel resistance and capacitance that causes a Wien bridge to null with the following component values. R_1=3.1kΩ C_1=5.2μF R_2=25kΩ f =2.5kHz R_4=100kΩ	**10**
10	An ac bridge with terminals ABCD has in Arm AB a resistance of 800Ω in parallel with a capacitor of 0.5μF, Arm BC – a resistance of 400Ω in series with a capacitor of 1μF, Arm CD – a resistance of 1000Ω, Arm DA – a pure resistance R. (a) Determine the value of frequency for which the bridge is balanced (b) Calculate the value of R required to produce balance.	**10**

Q. No.	MODULE 5: Unit 1	Marks
1	Discuss the operation of Cathode Ray Tube with diagram.	**10**
2	List and explain Cathode Ray Tube features.	**10**
3	Draw the block diagram of Oscilloscope and explain the functions of each block.	**10**
4	Explain Dual Beam CRO with block diagram.	**10**
5	Explain Dual Trace Oscilloscope with block diagram.	**10**
6	With a neat diagram, explain the Basic elements of storage mesh CRT.	**10**
7	Explain with a neat diagram, Storage mesh CRT.	**10**
8	Discuss the Display of stored charged pattern on a mesh-storage.	**10**
9	Differentiate between Dual Beam CRO and Dual Trace Oscilloscope.	**10**
10	Explain the different operating modes in Dual Trace Oscilloscope.	**10**
Q. No.	**MODULE 5: Unit 2**	**Marks**
1	With a neat block diagram, explain Conventional standard signal generator.	**10**
2	Explain Function Generator with block diagram.	**10**
3	Explain Random Noise Generator with block diagram.	**10**
4	Explain basic Wave Analyzer circuit.	**10**
5	Explain Heterodyne Wave Analyzer with block diagram.	**10**
6	Explain RF Spectrum Analyzer with block diagram.	**10**
7	Differentiate between basic Wave Analyzer and Heterodyne Wave Analyzer	**10**
8	Differentiate between Heterodyne Wave Analyzer and Spectrum Analyzer.	**10**
9	With a neat block diagram, explain RF heterodyne wave analyzer.	**10**
10	With a neat block diagram, explain Spectrum analyzer.	**10**